

Assignment 2

Q1. Create a program that reads student marks from a file (filename.txt). Display the names of students who have passed (marks > 45) and those who have failed. Count the total number of passed and failed students.

Concepts Used:

- File Handling
- if-else
- for loop
- Counters

Code:

```
#!/usr/bin/env python3
# Assignment2_file_handling.py

# Question 1
# Read student marks from a file (filename.txt)
# and display the names of students who have passed (marks > 45)
# and those who have failed. Count the total number of passed
# and failed students.

# Open the file
f = open('filename.txt', 'r')

# Read the file
data = f.readlines()

# Initialize counters
passed = 0
failed = 0

# Print the header
print("Passed Students")

# Loop through the data
for line in data:
    # Split the line into name and marks
    name, marks = line.split()

    # Convert marks to integer
    marks = int(marks)

    # Check if the student has passed
    if marks > 45:
        passed = passed + 1
        print(name)
    else:
        failed = failed + 1

# Print the total number of passed and failed students
print("Passed: ", passed)
print("Failed: ", failed)

# Close the file
f.close()

# Print the header
print("Failed Students")

# Loop through the data
for line in data:
```

```
print("Passed Students")

# Loop through the data
for line in data:
    # Split the line into name and marks
    name, marks = line.split()

    # Convert marks to integer
    marks = int(marks)

    # Check if the student has passed
    if marks > 45:
        passed = passed + 1
        print(name)
    else:
        failed = failed + 1

# Print the total number of passed and failed students
print("Passed: ", passed)
print("Failed: ", failed)

# Close the file
f.close()

# Print the header
print("Failed Students")

# Loop through the data
for line in data:
    # Split the line into name and marks
    name, marks = line.split()

    # Convert marks to integer
    marks = int(marks)

    # Check if the student has passed
    if marks > 45:
        passed = passed + 1
        print(name)
    else:
        failed = failed + 1

# Print the total number of passed and failed students
print("Passed: ", passed)
print("Failed: ", failed)

# Close the file
f.close()

# Print the header
print("Failed Students")

# Loop through the data
for line in data:
```

Output:

```
Run Assignment2_file_handling x

/usr/local/bin/python3.15 /Users/kirandeepsohi/Downloads/GENAI-PBEL/Assignment2_file_handling.py
Passed Students
Aa,
Bb,
Dd,
Ff,
Hh,

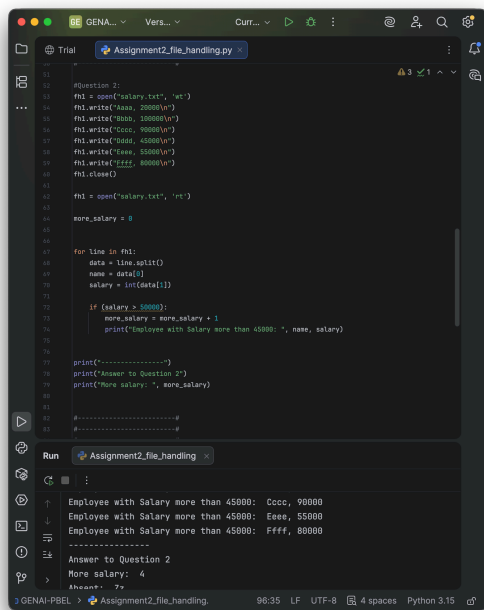
Failed Students
Cc,
Ee,
Gg,
Answer to Question 1
Passed: 5
Failed: 3
Employee with Salary more than 45000: Bbbb, 100000
Employee with Salary more than 45000: Cccc, 90000
Employee with Salary more than 45000: Eeee, 55000
Employee with Salary more than 45000: Ffff, 80000
-----
```

Q2. A file (salary.txt) contains employee names and salaries. Read the file and display only those employees whose salary is greater than ₹50,000. Also count how many employees satisfy this condition.

Concepts Used:

- File reading
- if statement
- Loop
- Counter

Code:



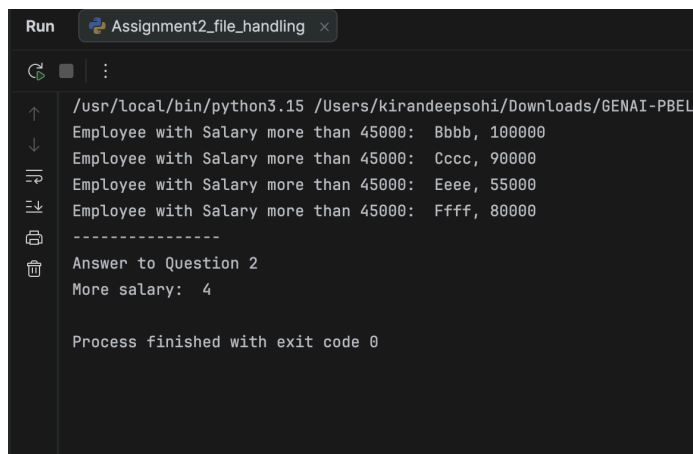
```
1 #Question 2:
2 fhl = open("salary.txt", "w")
3 fhl.write("Aaaa, 20000\n")
4 fhl.write("Bbbb, 100000\n")
5 fhl.write("Cccc, 90000\n")
6 fhl.write("Dddd, 45000\n")
7 fhl.write("Eeee, 55000\n")
8 fhl.write("Ffff, 80000\n")
9 fhl.close()
10
11 fhl = open("salary.txt", "r")
12
13 more_salary = 0
14
15
16 for line in fhl:
17     data = line.split()
18     name = data[0]
19     salary = int(data[1])
20
21     if (salary > 50000):
22         more_salary = more_salary + 1
23         print("Employee with Salary more than 45000: ", name, salary)
24
25 print("-----")
26 print("Answer to Question 2")
27 print("More salary: ", more_salary)
28
29 #-----#
30 #-----#
```

Run Assignment2_file_handling

Employee with Salary more than 45000: Cccc, 90000
Employee with Salary more than 45000: Eeee, 55000
Employee with Salary more than 45000: Ffff, 80000

Answer to Question 2
More salary: 4
Aheant+ 7>

Output:



```
Run Assignment2_file_handling x
/usr/local/bin/python3.15 /Users/kirandeepsohi/Downloads/GENAI-PBEL
Employee with Salary more than 45000: Bbbb, 100000
Employee with Salary more than 45000: Cccc, 90000
Employee with Salary more than 45000: Eeee, 55000
Employee with Salary more than 45000: Ffff, 80000
-----
Answer to Question 2
More salary: 4

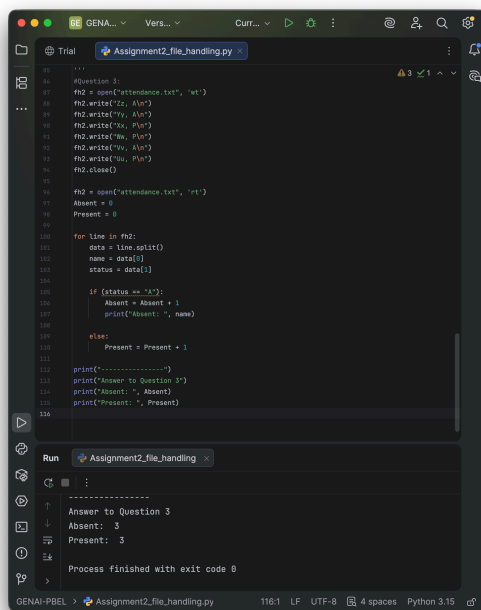
Process finished with exit code 0
```

Q3. A file (attendance.txt) contains the attendance status of students (Present/Absent). Read the file and display only the names of absent students. Also calculate the total number of present and absent students.

Concepts Used:

- File Handling
- if-else
- Loop
- Counting

Code:



```
1 #Question 3:
2 fh2 = open('attendance.txt', 'w')
3 fh2.write('Zz, Absent\n')
4 fh2.write('Yy, Absent\n')
5 fh2.write('Vv, Absent\n')
6 fh2.write('Ww, Present\n')
7 fh2.write('Uu, Present\n')
8 fh2.close()
9
10 fh2 = open('attendance.txt', 'r')
11 Absent = 0
12 Present = 0
13
14 for line in fh2:
15     data = line.split()
16     name = data[0]
17     status = data[1]
18
19     if (status == 'Absent'):
20         Absent = Absent + 1
21         print('Absent: ', name)
22
23     else:
24         Present = Present + 1
25
26 print('-----')
27 print('Answer to Question 3')
28 print('Absent: ', Absent)
29 print('Present: ', Present)
```

Run Assignment2_file_handling

Answer to Question 3
Absent: 3
Present: 3
Process finished with exit code 0

Output:

```
/usr/local/bin/python3.15 /Users/kirandeepsohi/Do
Absent: Zz,
Absent: Yy,
Absent: Vv,
-----
Answer to Question 3
Absent: 3
Present: 3

Process finished with exit code 0
|
```